

Artificial Intelligence

Viktor Dörfler, Senior Lecturer in Information & Knowledge Management, Department of Management Science, University of Strathclyde Business School, Glasgow, UK

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Email address: viktor.dorfler@strath.ac.uk (V. Dörfler)

Introduction

I have spent most of the last 20 years moving back and forth between two areas: As a practitioner I have developed AI software and used it to support decision makers. As a scholar I have been studying the human mind, focusing particularly on the highest level of expertise and on elusive phenomena, such as intuition and creativity. Naturally, I got interested whether AI can perform on par with the human mind, supersede it, and whether there is anything unique in the human mind that cannot possibly be achieved by AI as expressed through machines. Ultimately, my most important question always has been how to create a synergy between human mind and AI; in other words, I became interested in what is recently referred to as Augmented Intelligence.

Why is it useful or perhaps even necessary to have a chapter on AI in a creativity encyclopedia? The first, and perhaps, the most obvious reason is that many people today want to know whether we can ever expect AI Creativity (Amabile, 2019). Second, if there is a chance that AI can play a role in human creativity, it is worth exploring. The third reason, conceivably the least obvious one, and the most significant in my view, is that understanding AI (even in non-technical terms), can help creativity researchers understand human creativity better. The reason for this significance is that as we are trying to build AI, we must use what we know about the human mind, and in the process, we achieve greater clarity of the limitations of our understanding of the human mind. This is perhaps the most obvious in the case of creativity. For instance, we often intuitively classify an achievement as creative, and we do not think about explaining what makes it creative. However, if we compare such creative achievement with a similar or even superior achievement by AI, we do not necessarily want to label it as creative. This makes us think about what qualifies some achievement creative. In other words, if we want to put something into a machine, we need to put it into words, and in order to put something into words, we need to understand it. Therefore, my primary focus in this chapter is this third reason, although I will briefly address the other two as well.

The term artificial intelligence denotes both making intelligent machines as well as studying them. In line with the above argument, AI gurus, such as Herbert Simon, often noted that the field of AI includes studying the human mind. AI is loosely defined as machines that can accomplish tasks that humans would accomplish through thinking (through intelligence, as intelligence in AI is considered much broader than psychometric intelligence). It is interesting that the definition does not say anything about accomplishing such tasks the same manner as humans do; I will get back to this point below. To set the scene, I start with a brief overview of the history of AI.

A Brief History of AI

There are two distinct roots of AI, one is usually referred to as symbolic AI, the other connectionist AI. These two roots have come together, to a varying extent, in today's AI applications, but started as separate roots. Before the "AI-era" began, there was, however, one important work published where Alan Turing (1950), in his seminal paper, introduced the Turing test, machine learning, genetic algorithms, and reinforcement learning. In many ways we could say that we are still working our way through the challenges outlined by Turing.

The Symbolic Approach

According to the AI folklore, the history of AI began when Herbert Simon returned to the Carnegie Institute of Technology (today Carnegie-Mellon University) after the New Year in January 1956 for his class on Mathematical Models in the Social Sciences and told his students: "Over the Christmas holiday, Al Newell and I invented a thinking machine". One of the students in the class was Edward Feigenbaum (1992: 3), who recounts this sentence as the reason to become Simon's PhD student, and who was later to become the father of knowledge-based expert systems, one of the important areas of AI. During the same year, a conference was organized in the Dartmouth College, at which John McCarthy coined the term "Artificial Intelligence".

The Logic Theorist was the "thinking machine" that Newell and Simon created together with Cliff Shaw from RAND Co., the company that gave them access to a computer (a power-hungry IBM 701 with less than 20 kb memory, using unreliable vacuum tubes, costing millions of dollars). The Logic Theorist was supposed to try to find solutions to complex problems, such as proving mathematical theorems, when it is not even known if there is a solution. The Logic Theory Machine project was followed by the more ambitious General Problem Solver (GPS), which was supposed to extend this scope to any and all problem areas, such as deriving scientific rules from data or playing chess. This should have been achieved by following the steps of human problem solving, and replicate achieving the outcomes (i.e. solutions to the problems). The idea was that if we do this across a number of problem areas, we could develop AI that

could apply the general principles of reasoning, and thus would not only work within a narrow well-defined scope, but across various domains. This type of AI is also referred to as “wide-AI” or, more recently, Artificial General Intelligence (AGI).

Moving to Berkeley, away from the direct influence of Simon and Newell, Feigenbaum got more and more interested in creating a machine of superb performance, possibly superior to human performance. Instead of focusing on the general principles of reasoning, he focused on the role of specialist knowledge at a high level of expertise. His dream was to build a super-intelligent machine. As he realized that the previous AI solutions were good at deductive but not so good at inductive reasoning, Feigenbaum was looking for a problem in which hypothesis formation could be captured by inducing internal models of external environments. Partnering with the Nobel Laureate Joshua Lederberg, he ended up with the problem of inducing topology of molecules based on their mass spectra. The DENDRAL project, as it was called, lasted for many years, involved further subject area experts, but at the end, the program was performing at the level of the top experts. This was the first knowledge-based expert system. A system being “knowledge-based” refers to the internal model called “knowledge representation” which was stored in the form of a “knowledge base”, while the term “expert system” signified that expert knowledge was being modeled.

Both, the GPS (and the previous Logic Theorist) and the DENDRAL relied on a process called “knowledge acquisition”, through which human thinking about a problem was captured in the form of logical “if... then” rules. As such rules can be implemented as symbol manipulations, such systems became known as symbolic logic, or symbolic AI. There was, however, a great divide between the two directions in symbolic AI. One direction was to attempt building a domain-independent AGI, through capturing the generic process of problem solving. The other direction was to build expert systems (or knowledge-based systems or knowledge-based expert systems) in a well-defined narrow area by modeling subject-specific knowledge of experts. Knowledge bases typically comprised a few hundred to a few thousand “if... then” rules, they rarely reached the range of tens of thousands (and this was inevitably only possible by involving multiple experts).

Knowledge-based expert systems dominated the AI landscape until the mid-1980s, producing a large number of very successful implementations in a wide range of areas from science through manufacturing to medical treatments. At the same time, the AGI approach did not produce any implementations that would successfully work across multiple domains, but some implementations were successful in narrow domains.

The Connectionist Approach

While the symbolic approach to AI was dominating the early days of AI, the connectionist approach has been developed earlier. The first Artificial Neural Network (ANN) implementation was done by Marvin Minsky and Dean Edmonds, as part of their graduate work at Princeton in 1951. The purpose of these ANNs was to demonstrate that any mathematical function could be calculated by a suitable network of artificial neurons; therefore, this was not what would today be regarded as AI, but might be regarded as “pre-AI”.

ANN, however, became somewhat dormant until the mid-1980's, when it had a big comeback. There were several reasons for this comeback. The AGI approach did not produce any solutions that would work across domains, expert systems were confined to their narrow areas and took extremely long times to develop (often becoming obsolete once the problem was solved), and computers became sufficiently fast to cope with ANN of a useful size. ANN are today perhaps the most heard of forms of AI, in fact, ANN seems to have monopolized much of the AI area, including the concept of machine learning. Therefore here I provide a very much simplified but in-principle correct description of what ANN are and how they work.

Any ANN consists of three layers of artificial neurons. There is an input layer, that receives some kind of signal (stimulus)—a stimulus can be more or less anything we can convert into an electronic format. Then there is a hidden layer, which performs some sort of black-box operation between the input and the output. Finally, the output layer produces a response to the stimulus received by the input layer. Nowadays we often hear terms like “deep ANN” or “deep learning”, which sound mystical and exciting; in fact, these simply mean that there is more than one layer of artificial neurons in the hidden layer. Computationally this makes a great deal of difference, but logically the concepts are simple. The artificial neurons are connected to each other across the layers, like synapses in the human brain, hence the term connectionist approach. Each connection has an initial weight, so as the stimulus “ignites” the system, the signal propagates toward the output neurons resulting in a response. The response is compared to what was expected, the weights are adjusted, and the process is repeated iteratively. As it is easy to process a large number of learning samples (stimulus and the desired response), the ANN can relatively quickly adjust to producing the desired response. In other words, ANN learns from a large number of learning samples how to reproduce the statistical frequency of those learning samples.

So the logic is simple, but then comes a twist. As the artificial neurons were modeled after the biological neurons, more precisely after our knowledge of the neurons at the time, the ANN, supposedly, resembles the human brain. Therefore, so the argument goes, if we put together a sufficiently large ANN, it should think. Furthermore, as the digital neurons are faster than the biological ones, these machines should be, following this line of thought, smarter than humans are. Although this logic may appear sound at the first sight, it is heavily based on assumptions and beliefs, and also seems quite unrealistic for the present times – at least to me. Let me unpack this assertion.

My first point is about the size of ANN. The human brain is taken to consist of about 100 billion neurons, with around 7000 connections on average, resulting in some 1000 trillion (10^{15}) synapses in total. The largest ANN I have seen consisted of a few dozen artificial neurons, although I know that there are some in the hundreds. It is not public knowledge how many neurons the most advanced ANNs, such as the one in DeepMind, comprise today but I would expect a few thousand. In these exceptionally advanced systems, an artificial neuron perhaps may have a few hundred connections. This means a few hundred thousand synapses in total, suggesting that ANN is only about 10 orders of magnitude short of the human brain's network. True, in the last 70 years ANN architects managed to

achieve an increase of 4 orders of magnitude, so another 10 does not sound like science fiction. But it is surely reaching the level of complicatedness (rather than complexity, see later) which cannot be grasped by the architect anymore; the only way to analyze such huge AI is using AI. In other words, I am hesitant to believe it is only a matter of time until we produce functioning ANN of a similar size as that of the human brain.

My second point is about the structure of ANN. In such a network the connections always go forward, that is a neuron from any layer only connect to the ones from the next layer (in some networks there can be within-layer connections as well). We do not really know if there are any circular connections in our brains, but there may well be. Furthermore, I am not even sure that we should think in “layers” about the brain. In other words, while structurally ANN resembles the brain in being complicated, a closer look at the structure also reveals dissimilarities.

The third point concerns the functional aspect of ANN. The functioning of artificial neurons reflects what we knew about biological neurons around the mid-20th century—that neurons are either “firing” or not, resembling the digital computers. Today, it seems that at least a small subset of neurons (maybe only a few million?) displays a more complex behavior, more akin to analog computers. This means that functionally ANN is rather limited in comparison with the human brain.

If ANN architects managed to overcome all the difficulties of the previous three points, they could construct something like an artificial brain. Some enthusiasts emphasize that as the artificial neurons are faster than the biological ones, the artificial brain should be better, or at least faster, than the biological one. However, here we face another assumption barrier: that the brain, if structurally, dynamically, and functionally adequate, would produce the mind. It may be that many find this assumption plausible, but it is just a belief. However, what science knows so far is that the brain and the mind are somehow connected, and particular thinking processes can be associated with neural activities in particular areas of the brain. We do not really know whether the brain produces the mind, the mind produces the brain, they co-produce each other, or the brain and the mind are produced separately and linked. The typical computer science analogy for cognition is that the brain is the hardware and the mind is the software—although the software is running on the hardware, hardware has never produced software.

Finally, recent research also suggests that the brain is not the only part of our biological setup that participates in our cognitive system; our endocrine systems also seem to be part of it, and I would not be surprised if we found in the coming years that our whole body is, meaning that the mind is fully “embodied”. In other words, if an artificial brain was produced, reasonably approximating the human brain in size, structure, and functioning, this does not imply that it will produce an artificial mind.

AI Paradigms

I have noted above two divides in AI. The first divide is between the symbolic and connectionist approaches, a divide which is technologically fundamental. There are further tools and implementations that belong in the AI landscape, such as fuzzy logic and genetic algorithms. I have not gone into the details of these here, as they do not offer different approaches to AI. Fuzzy logic is usually used in a connectionist way, in combination with ANN or on its own. Genetic algorithms are well aligned with ANN, and they could be used in symbolic AI as well—although I have only seen examples of this in computer games. The second divide is between artificial general intelligence (AGI), aimed at wide application area with less concern on current performance, and expert systems, aimed at high performance in a narrow area. In this section, I introduce a third divide, which is somewhat like an extreme version of the AGI versus expert system divide; these are the paradigms of AI: strong AI versus weak AI (referring to strong versus weak assumptions).

Strong AI means creating a thinking machine that equals humans in every cognitive aspect. According to the proponents of strong AI, these machines were not only expected to think, and deliver performance that is on par with human, they were also expected to think the same way as humans do. This kind of AI is expected to awaken to self-consciousness, and have intentions, emotions and sensitivity. In principle, there are two ways to achieve strong AI. The first way is to attain a complete understanding of the human mind and then the ability to replicate it artificially. Unfortunately, or luckily, depending on where you stand, we are nowhere near to being able to understand the human mind. If we take complexity theory seriously, we have to accept that we cannot ever achieve this, as no system can understand another system that is at least as complex as itself. The second way to achieving strong AI is to admit that the human mind cannot be fully understood, at least for now, but maintain the belief that it is possible to create an artificial brain that will then produce an artificial mind, which is like the human mind as we know it. Based on everything stated above, and on what most AI experts say, having a human-level and human-mode thinking machine cannot happen in the near future. However, there are a few enthusiasts, some of them the greatest minds of AI, such as **Simon (1977: 6)**, who said:

“I believe that in our time computers will be able to perform any cognitive task that a person can perform. I believe that computers already can read, think, learn, create ...”

Those who subscribe to the weak AI paradigm, believe that we do not, at least currently, understand the human mind well enough to reproduce it. Perhaps even more importantly, they want to align their efforts with a purpose that sounds less grandiose, but is aimed at producing useful and meaningful results right now. So they want to build specialist applications, such as expert systems, that work well in a narrow predefined domain, and only performance is of interest, it does not matter if the machine achieves it in the same way as a human would. My personal beliefs are closer to this approach and, as a practitioner, it is certainly my preference to produce something that works right now. I always tell my students a metaphoric example: “When they wanted to go somewhere, our great-grandfathers called a carriage. Today we call a taxi. The taxi is not an artificial horse, it works on completely different principles, but can do the same job of taking us where we want to go, in a different way, perhaps even better (e.g. more conveniently and faster). In order to make the taxi, we did not

need to understand everything about the biology (and perhaps even psychology?) of horses, we had to understand the internal combustion engine and the wheel. Of course, the taxi cannot substitute the horse in every task, for instance we cannot play polo on it. But it can take us where we want to go”. Similarly, today we understand a lot more about computers than we understand about the human mind. And we can make useful tools using computers, using AI. AI is far better at many things than humans are, often these are things that humans do not find particularly interesting anyway – and it cannot do some things, at least for now, that humans find easy. Then, again, this is also true for a seeding machine or a coffee mill.

To be fair to those in the strong AI paradigm, they do produce things that work right now. The stories on the websites of DeepMind or Watson show a number of projects that are incredible, useful as well as meaningful, while others are interesting and funny. Perhaps it would also be fair to say that DeepMind and Watson transcend the strong-weak AI paradigmatic divide.

What AI Still Can’t Do—and What It Does Well?

So how do we know that a computer is thinking? How do we know if it should be considered (artificially) intelligent? The first criterion is the Turing test (1950), which is still usually referred to as a qualifier. The essence of the Turing test is that if we interact with an entity and we cannot figure out whether we are talking to a person or to a machine, and it is in fact a machine, that machine should qualify as intelligent. On first sight this sounds convincing—if the machine is not really thinking, we should be able to catch it out. So has any machine (computer program) passed the Turing test so far? Yes, quite a few did, from ELIZA through a number of chatbots, to Eugene Goostman, the simulated 13-year-old Ukrainian boy, and an AI course moderator at MIT. Instead of getting into the details whether these instances of passing the test were legitimate, let me ask different questions. So what if a machine passed the Turing test? Would that really mean that it was thinking in the sense we attribute this verb to humans? No, I do not think so. The programs that passed the Turing test were designed with this purpose, not with the purpose to think.

Perhaps the most convincing argument against the Turing test comes from John Searle (1998: 11)—that became known as the “Chinese Room Argument”. The essence of the argument is the following: assume that I do not speak Chinese, and I am in a room full of rule-books, receiving messages in the form of Chinese symbols and looking up responses to those messages in the rule-books. If the rule-books are good, those outside the room, receiving the responses, would believe that the person responding understands Chinese. And they would be wrong, I did not miraculously learn Chinese. The same way, passing the Turing test does not prove that the computer thinks, only that the program is good.

Minsky (2006) reposts to Searle by reversing the Chinese Room Argument, creating a hypothetical “zombie machine”. The zombie machine is seemingly identical to the human and, if it injured a leg, it would complain about the pain (as it was programmed to do so). The response to the pain would be indistinguishable from a human with a pain in the leg. If this was done, could we claim that the zombie machine’s leg does not hurt—that it does not feel? But the most brilliant part is the final turn: if we cannot distinguish the zombie machine from a human being, can we claim that the human being feels?

Of course, none of the above proves that the machine does not think, the arguments only falsify the validity of the evidence that it does. So, we are left to our beliefs as there are no conclusive proofs. Another important thing that we can learn from the above is that we do not have a way to directly measure knowledge or thinking, what we can do is make inferences from the performance. This is not a unique situation in the world of science, all the work done in quantum physics is similar, elementary particles cannot be put under a microscope and it is impossible to measure their weight or speed or similar. What physicists do instead is to observe interactions of elementary particles and infer their characteristics. Physicists come up with models, speculatively, conduct experiments in which they measure some effects, and from these they infer various characteristics and check whether these are consistent with their speculative models. That is probably the reason that there is no single model accepted by all quantum physicists. Many different models can be consistent with the same results, depending on our inferences.

So, instead of trying to prove that AI can or cannot think, it is perhaps more useful to look at the performance of AI and the related claims. In 1957 Herbert Simon announced that the chess world champion will be an AI machine within 10 years. However, it was only in 1997, four times longer than Simon predicted, that Deep Blue defeated Garry Kasparov, who is possibly the best (human) chess player of all times. Since then we had AlphaGo defeating the world champion Lee Sedol as well as all the other Go grandmasters, AlphaGo Zero winning consistently against AlphaGo, and AlphaZero beating not only the previous two but also the best computers in chess and shogi. Let me unpack how AI achieved this extraordinary performance and what it means.

There are four different “pure” ways for AI to become “smart”. The first is to hardcode a program specific to the chosen performance area. These are “purpose-built programs”, such as Deep Blue which was programmed to play chess. The second way could be termed “process capturing”; it characterizes the previously mentioned Logic Theorist and the subsequent General Problem Solver. The essence of this approach is to record the steps of problem solving by human experts using the “thinking aloud” technique. The third approach is “knowledge representation”, which is primarily used by knowledge-based expert systems. Knowledge engineers build knowledge bases from knowledge acquired from domain experts, while the processes of how these knowledge bases are used, are programmed. Finally, the fourth approach, “machine learning”, is to let AI learn with or without humans interfering with the process. This approach is predominantly associated with ANN, but all forms of machine learning belong here, including those in the symbolic AI realm (e.g. Quinlan, 1986). Of course, apart from these pure approaches there are also various mixed approaches within and across these categories.

Each of the above approaches has at least one of the following two characteristics: they require *explicit knowledge* or they follow a *reinforcement learning process*. Both of these impose severe limitations on what we can expect AI to do. In fact, these limitations should

be sufficient in themselves to convince most people that AI, as approached today, can never think and learn in the sense of human thinking and learning. This also suggests that AI Creativity, in the human mode, is impossible. This is explored at a greater depth in the next section; here I want to examine the broad implications of these two characteristics.

The first characteristic limits human knowledge to explicit knowledge. While it is widely known that humans also have, and use, tacit knowledge, **Polányi (1966: 7)** goes much further claiming that there is no knowledge without tacit knowledge:

“While tacit knowledge can be possessed by itself, explicit knowledge must rely on being tacitly understood and applied. Hence all knowledge is either tacit or rooted in tacit knowledge. A wholly explicit knowledge is unthinkable.”

Considering that much of what we know about knowledge today originates from Polányi, we must take seriously his assertion. AI that uses any form of knowledge representation, hardcoded, in the form of knowledge bases or as embedded problem solving steps, is limited to explicit knowledge. This explicit knowledge is always used in the process of formal logic, and even Simon, who was extremely optimistic about the prospects of AI, warned that formal logic is only one of the instruments of reasoning. Knowledge-based expert systems respond to this limitation by focusing on a narrow domain and fine-tuning the knowledge bases together with the domain experts, until it reflects their judgments. In the process, occasionally, something in the tacit knowledge of the experts becomes explicit, that is, something is spelled out that the experts already knew, only they did not know that they knew it. This rare event is called “knowledge discovery” and is often considered to be the most valuable aspect of the knowledge engineering process.

Limiting learning to reinforcement learning is at least as significant a departure from human learning as limiting knowledge to explicit knowledge. There was a period in the history of psychology, called behaviorist psychology, when reinforcement learning was considered to be the only way of learning. While it was therefore sensible for Turing to suggest reinforcement learning in 1950, psychology has come a long way since its “dark ages”, when it denied its own topic of research considering the mind a black box. At the heart of the reinforcement learning approach is the stimulus-response (S-R) model; the entity (human, animal, machine) receives a stimulus and provides a response. We reward the preferred response and/or punish the non-preferred ones. Sooner or later, the responses will align with our preferences, and this is considered learning. Of course, there are a few things that humans learn this way, for example a child will quickly learn not to touch a hot stove. However, we also learn from stories, get inspired, are talented for some things and not for others, commit to master-apprentice relationships, etc. Reinforcement learning covers only a tiny part of our learning.

Both of these issues, limiting knowledge to explicit knowledge and limiting learning to reinforcement learning, are captured in the concept of “common sense”, contrasted to specialist knowledge. As **Minsky (1988)** said, it seems to be easier to create AI that performs at an expert level in a narrow domain than to make a program which could play children's games or bring in the newspaper from the front door to the breakfast table. According to Minsky, common sense is an immensely complex system of rules, and exceptions, practical ideas, dispositions and tendencies, balances and checks. It appears simple and natural as we have lost touch with how we learned them—and we know almost nothing about how common sense is acquired. Interestingly, most researchers from strong and weak AI paradigms agree that the missing link to creating a thinking machine, and by implication AI Creativity, is the model of common sense. The difference is that according to the strong AI community it is only a matter of time before a model of common sense is obtained, while according to the weak AI group it is impossible. There were several attempts to develop artificial common sense, among others the “CYC knowledge base” (by Cycorp), which is probably the largest knowledge base ever produced, comprising “millions of facts, beliefs, and other bits of knowledge” (**Feigenbaum, 1992: 17**). All these attempts have remained futile so far and none showed any promising progress toward modeling common sense. Perhaps, it would be sensible to ask whether there is any meaningful concept of intelligence (human or artificial) at all without common sense.

The above is a sufficiently convincing argument that AI does not think or learn the way humans do. However, this does not necessarily mean that AI does not think or learn in ways that are different or only partially overlapping with the human ways. As I indicated previously, to deliver a particular performance, it is not necessary to do it as humans do – and we can see that AI can beat us in areas like the game of Go. All the success stories, however, are in relatively narrow domains. With rare exception, such as DeepMind and AlphaZero, this means one single problem area. DeepMind is usually represented as a wide-AI. However, it does not cover the whole spectrum of human problem solving, it only plays various computer games designed for Atari computers (an early PC of the 1980's), while AlphaZero plays Go, chess and shogi. Although it is a significant step forward that DeepMind (which is also at the core of AlphaZero) can cover more than one domain in a single system, playing the different games is very far from covering a full range of human problem solving, such as leading an organization, writing a poem, nursing a patient, or enjoying a football game (as a fan). In short, there is convincing evidence that artificially reproducing thinking and learning (and by implication creativity) in a human mode, is impossible, at least for the time being. In the next section, I explore how AI does in the area of creativity, which is traditionally considered a particularly human endeavor.

Creativity and AI?

Can AI be creative? If we reverted to Polányi's above assertion, the short answer would be a sound “no”, because, as the renowned philosopher of science, Karl **Popper (1968: 8)** put it:

“[T]here is no such thing as a logical method of having ideas, or a logical reconstruction of this process. My view may be expressed by saying that every discovery contains «an irrational element», or «a creative intuition», in Bergson's sense.”

In other words, we know that intuitive insights (Dörfler and Ackermann, 2012; Dörfler and Eden, 2019) play a significant role in human creativity, they appear to be a necessary ingredient, and they belong in the realm of tacit knowledge, therefore AI Creativity, in a human mode, is impossible. However, it is worth considering whether there can be AI Creativity in a way that is different from how humans do it. In order to explore this further, it is useful to reconsider the AI track record so far and how these relate to the definition of creativity.

There are many claims about how AI will create at the finest level of human creative performance, such as writing a Shakespeare-level prose as suggested by Minsky (Amabile, 2019), and according to some this has already happened. One of the earliest acclaimed creative achievements of AI is the Logic Theorist (LT) coming up with a proof for one of the theorems in the Principia Mathematica that is according to Alfred North Whitehead “more elegant” than the one he and Bertrand Russell provided. This was definitely something new and useful—the two criteria of creativity. However, if we use the original definition of creativity by Teresa Amabile, as a novel and useful idea, it has a hidden third requirement: it also has to be an idea. While the details of how the LT did this requires some technical knowledge, it is easy to see that the process is transparent and fully explicit—contrary to how we see the human creative process. Can it help if we ask an appropriate group of judges to evaluate whether an AI outcome is creative and to which extent? The inadequacy of the Turing test suggests that this is not the case. To elaborate this, let us look at the two great game achievements, namely Deep Blue defeating Kasparov and AlphaGo defeating Lee Sedol.

In both cases there was a master-stroke, that shocked the human players and was highly praised by the chess and Go experts respectively. In the case of Deep Blue, a programmer later on said that there was a glitch in the system, caused by a bug that was subsequently fixed, which did not allow Deep Blue to choose one of the evaluated steps, so it went ahead with a random step instead. As in the Chinese Room argument, it was Kasparov who put the sense into this step. AlphaGo had a move against Lee Sedol that corresponds to all the rules but was never made by a human Go master. The reason is that Go has an enormous number of possibilities (and therefore it is more complicated, although less complex than chess), and the tradition evolved toward playing line 3 or 4 in particular situations and nobody ever played line 5. However, as AlphaGo, during its “training period”, played some 300 billion games (which would be many grandmaster lifetimes) against itself, it built a database of moves, and the surprising move had a higher probability for a win in that particular situation. A first look at these stories suggests that there was no creativity, in the human sense, demonstrated by AI in either case. However, if that was all we could learn from this case, it would not be worth going into such detail—there is much more.

If we relax the definition of creativity to include “ideas, problem solutions, or other outputs” (Amabile, 2019: 3), the two moves, as well as the above mentioned mathematical proof provided by LT, could qualify. I would be still hesitant to consider these achievements creative, as it is fairly visible how they were achieved, and looking up maximum probability from a big table intuitively does not appeal as creative. If we think about this, the move by AlphaGo was new to the human Go masters, but it was not new for AlphaGo—it must have made it many times, if it had high probability for a win. Was it a new move when AlphaGo made it the first time? It probably was, but randomness seems to have more to do with this than creativity.

Nevertheless, these examples do give us an important glimpse into how AI can play immensely useful role in human creativity. AI is really good at quickly analyzing large amounts of data, identifying patterns in vast sets of facts, things that require precision. That is why Thomas Davenport (2018: 44) says that AI is “analytics on steroids”. Humans are not very good at these, and even if we managed to do it, it would take far too long. What AI is not so good at, is judging which pattern is, or may potentially be, significant in the situation at hand – but human experts excel at this. For example, if an oncologist and a chemist are working with some chemicals trying to achieve an effective cancer treatment, AI may produce an incredible number of combinations of chemicals at hand, but it cannot judge which resulting compounds may be of interest, beyond meeting predefined specifications. However, the oncologist and the chemist may figure out, looking at some of the characteristics of the new compounds produced by AI, whether there are some new (unexpected) combinations which may be useful for curing cancer. Furthermore, human experts may also notice that some of the resulting compounds may be useful for treating kidney stones, for instance. The reason that AI could not pick up on these is that the solution is in a different realm than the original problem. The knowledge of the two doctors was necessary for figuring out the significance of the new pattern uncovered by AI. Similarly, designers in fashion, furniture, automotive and other industries use AI today for coming up with potential new models. While AI can deliver new models (by rearranging existing patterns), it is the designer who chooses which one is aesthetically appealing. Therefore AI can provide pre-processing in the sense of identifying and organizing patterns, and so can add immense value to the human creative process.

There is an additional perhaps less trivial value that AI can add. High-level human creativity is closely associated with domain (discipline or problem area) expertise. Such high level of expertise takes time to develop. And, like all the Go grandmasters in the world, experts often lock themselves up “inside the box”, imposing unnecessary but conventional limitations. In fact, the triumph of AlphaGo revitalized the world of Go: human Go grandmasters are reporting new moves, new strategies, increased openness, and going beyond the approaches of the traditional schools. Consequently, although it may sound initially surprising, AI can help us think “outside the box” in the best sense of the term.

Concluding Remarks

In the prologue of his book called *The Emperor's New Mind* (paraphrasing Hans Christian Andersen), Roger Penrose (1989) introduces a mythical supercomputer Ultronic, which was set up in the Grand Auditorium, and the citizens were invited to ask any question. As the computer knew everything, it should be able to answer any question. People felt “bashful, afraid to seem stupid”, intimidated by such a

super-smart entity, so nobody dared to ask any question ... until the thirteen year old Adam dared... The story is concluded in the Epilogue of the book, Adam asked the super-smart computer what did it "... FEEL LIKE", to which Ultronic responded that it didn't see what and couldn't even understand what Adam was getting at and then everyone was laughing. The supercomputer with 1017 logical units, which was supposedly making it more intelligent than the combined intelligence of the whole country, could not handle such a tough question.

In this chapter, I have provided a personal view of AI in the context of creativity. I have tried to paint a rich picture of AI, not aiming at comprehensiveness or immaculate precision of the details, but rather at intuitive understanding of the essence. However, I also tried to be faithful to what can be known about AI, bringing it closer to the reader. Herbert **Simon (1995: 508)** said that it was time to stop philosophizing and to focus on empirical (and particularly experimental) testing. While I agree with Simon about the value of empirical work, I think that philosophizing about AI was never as timely as it is now, in order to delineate beliefs from facts and to demystify what AI can actually do. Unless we do this, AI is in the danger of becoming a brilliant solution in search of an appropriate problem (as Theodore Roszak said about computers in general), rather than a brilliant solution to a range of problems in a variety of areas, including creativity.

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